RESEARCH CASE STUDIES

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Three projects from aerospace, healthcare, and space research. Each one pairs qualitative and quantitative methods, and each page says what the research changed.

1 Turning a cognitive task analysis into design questions engineers could test

Lockheed Martin, Future Vertical Lift, 2023 – 2026

Methods: design framework built on an existing cognitive task analysis, stakeholder working sessions, usability testing, experiment design

Outcome: three recommendations adopted into the engineering design process, and a UI overhaul informed by the findings

2 Finding where clinical workflows break, and fixing them

Orlando Health Imaging Centers, 2020 – 2022

Methods: operational data analysis, task analysis, field observation, diary study, staff surveys, focus groups, usability testing

Outcome: 60% fewer errors and 40% higher departmental productivity in under six months

3 Measuring how teams think without interrupting them

U.S. Department of Defense, NASA, Minerva Research Initiative, 2023 – 2025

Methods: collaborative problem-solving studies, task performance measures, communication coding with inter-rater reliability checks, wearable and environmental sensors, interviews, surveys, statistical analysis

Outcome: a team effort toward unobtrusive measures of individual and team cognitive states. Results are restricted.

CASE STUDY 1 OF 3

Turning a cognitive task analysis into design questions engineers could test

Externalized and Extended Cognition: Cognitive Offloading for Manned-Unmanned Teaming
Lockheed Martin, Future Vertical Lift, 2023 – 2026. Role: Research Lead

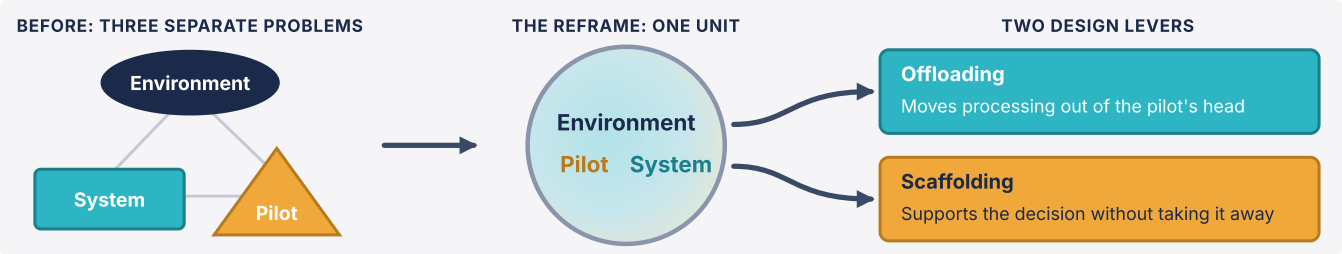
3 of 3 design recommendations adopted into Lockheed's engineering design process	**UI overhaul** of an active operations system, informed by the findings	**HFES 2024** first-author paper and presentation, doi.org/10.1177/10711813241275506

The problem as it was handed to me

Lockheed Martin needed to understand how to reduce pilot cognitive workload in next-generation manned-unmanned aviation, the Future Vertical Lift program. A detailed cognitive task analysis of the operational environment already existed. What was missing was a bridge from what pilots need cognitively to concrete, testable design questions that engineers could act on.

The choice I made, and why

I could have treated workload as one thing to measure and minimize. Instead I applied a cognitive science framework, 4E cognition, which treats cognition as extended, embedded, enactive, and embodied. It let me ask a more useful question. Not only how much workload there is, but where the cognitive processing lives: in the pilot, in the system, or in the interaction between them. That pointed to two separable mechanisms that engineers could design around independently.



How I worked with the engineers

I facilitated working sessions with Lockheed and partner engineering teams to align on objectives and turn their goals into research questions. I also ran usability testing on an operations monitoring and information management system.

What I produced

Three recommendations that turn the framework into testable design questions, and the experiments to test them.

1. **Look at the whole timeline.** Analyze workload across planning, the mission, and the debrief, since each one shapes the next.
2. **Vary how available offloading is.** Test it as required, optional, and unavailable, to see how that choice affects cognitive efficiency when tasks are disrupted.
3. **Study the interactions.** Examine how types of workload and offloading techniques combine, and what that does to mission performance.

What's still open

The hardest part was not the theory. It was translating it into something an engineer outside cognitive science could act on without oversimplifying it. Next time I would ask engineers for feedback on the recommendations earlier, before finalizing the experimental designs, to catch translation gaps sooner.

Implementation details are confidential and omitted.

CASE STUDY 2 OF 3

Finding where clinical workflows break, and fixing them

Quality Assurance Program and Epic EHR launch, Orlando Health Imaging Centers, 2020 – 2022
Role: Supervisor, Process Improvement Coordinator

60% fewer errors, in under six months	**40%** higher departmental productivity	**10 to 25** staff in the research, across four roles

The problem as it was handed to me

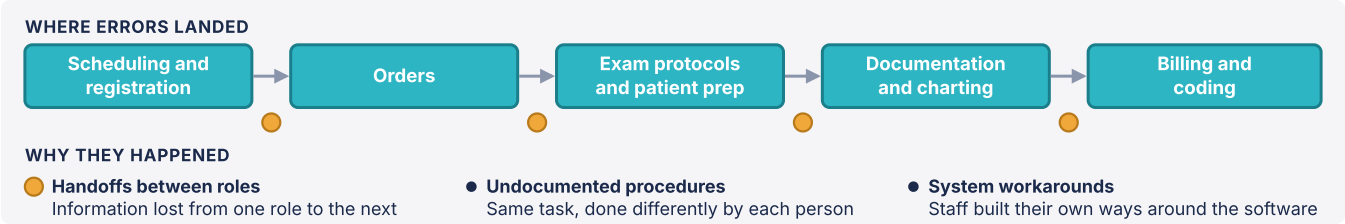
Errors were showing up at every stage of the imaging centers' work. The operational data could tell me where errors landed, but not why they happened, and without the why, any fix would have been a guess.

The choice I made, and why

I paired the numbers with time spent watching the work. Each method covered something the others could not.

METHOD	WHAT IT WAS FOR
Operational data analysis	Compared error and productivity data against internal and external benchmarks to locate the risk points worth studying. Done in Excel.
Task analysis	Mapped clinical and non-clinical workflows step by step, so each error could be tied to a specific point in the work.
Field observation	Watched front desk and registration staff, schedulers, technologists, and supervisors do the work in place.
Diary study	Ran for over a month, to capture problems that recur over time and would not show up in a single visit.
Surveys and focus groups	Collected workflow and process feedback and training needs from staff.

What I found



What changed

I defined performance metrics and led the process redesign. That meant standardized workflows with written guides, staff retraining, staffing and schedule changes, and ongoing audits with regular reporting on productivity, error, and revenue metrics. I presented the findings to stakeholders to build support for each change.

Epic EHR launch

When the organization moved to Epic, I managed training, built the launch reference resources, and ran usability testing on the EHR.

What's still open

The results are a before and after comparison. The changes went in together, so I can't say how much each one contributed on its own. With more time I would have staged them to find out.

Figures are aggregate operational results. No patient data is included.

CASE STUDY 3 OF 3

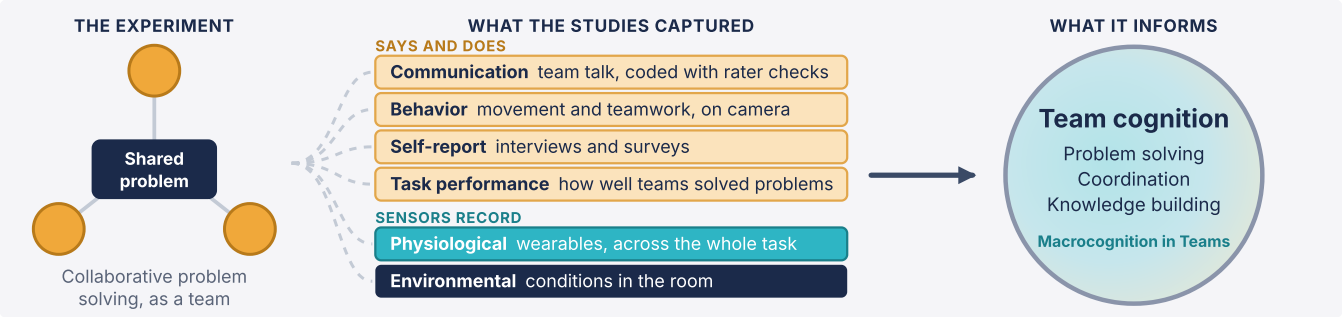
Measuring how teams think without interrupting them

Understanding Cognition in Support of Future Space Missions
U.S. Department of Defense, NASA, Minerva Research Initiative, 2023 – 2025
Role: Research Staff, Experiment Coordinator

> This project is unpublished and under Department of Defense data restrictions, so this page covers the methods and my role, not the results. The project wound down in 2025, after the Department of Defense ended the Minerva Research Initiative.

The problem

Long-duration space missions need a way to monitor the cognitive state of individuals and of the team as a whole. Our team set out to develop unobtrusive measures of team functioning. The studies put teams through collaborative problem-solving tasks and captured how they communicated, coordinated, and performed, alongside physiological and environmental sensor data.



What I did

METHOD	MY PART
Sensors	Helped assemble the team's sensor suite, and set up, ran, and troubleshot Empatica devices with participants during sessions.
Communication coding	Built and ran the behavioral coding and transcription training, coded team communication data, and calibrated the coding schemes across raters using inter-rater reliability checks.
Interviews and surveys	Ran participant interviews and surveys as part of the studies.
Statistical analysis	Ran my own analyses of the study data in R and Python.

Running the studies

As Experiment Coordinator I handled the IRB protocols, screened, recruited, and scheduled participants, trained and calibrated research assistants so sessions ran the same way every time, and coordinated with partner labs at other universities.

What the team produced, and my part

I helped develop the team's unobtrusive multi-sensor measures of individual and team cognitive states, and contributed to its workflow and environment design recommendations for long-duration missions with human and AI (onboard system) team members.

What carries over

The method is the part that travels. This project paired what sensors record with what people say and do, which is the same pairing of behavioral data and direct research that product teams rely on.